

Testing your submission

The primary intended use of releasing this script this late in the quarter is to help you test your homework submission in as close to an environment as I'm testing as I can come up with. On Moore, you should do the following:

1. Unzip your submission directory
2. If you are not providing your own library/Makefile, copy the Makefile for each language (starting from the language you're submitting for) into the respective directory a. These are the same Makefiles we originally provided, but they are included here in case you changed your Makefiles b. If you are not using external libraries, please use the Makefile for each language provided in this zip folder c. You will need to modify the Makefile so that the path to PEGTL points to your lib directory (I have a script that points to the lib directory in the test folder so I don't have to copy PEGTL 30 times)
3. Build each language such that there is a binary in the respective bin/ directory using just the `make` command.
4. Run the `test_compiler.py` script with the as follows from the language you are testing (e.g. in `your_submission/L3/`):

```
python3 PATH/TO/test_compiler.py .. LANGUAGE PATH/TO/TESTS . PATH/TO/runtime.c
```

For example, when testing the L3 language with a test suite and lib two levels above your language, I would run:

```
python3 ../../scripts/test_compiler.py .. L3 ../../tests/ . ../../lib/runtime.c
```

If things are taking awhile or you run into problems, consider adding a verbose flag and an output directory that can show you more details of your errors for each test in `.tmp` files:

```
python3 PATH/TO/test_compiler.py [other flags] -v -o output
```

If you're failing a test due to a timeout, feel free to up the limit (I'll permit a few slow tests this quarter without complaint):

```
python3 PATH/TO/test_compiler.py [other flags] -t 1000
```

For testing liveness, interference, and spilling, the process is similar (using e.g. `test_liveness.py`), but with a small difference. Running from your language folder, all you need to provide is your binary location and the test suite:

```
python3 PATH/TO/test_liveness.py bin/L2 PATH/TO/TESTS
```

Finally, note that you are welcome to use these python scripts to test rather than the provided shell scripts, you can just update your makefile to test with the python script instead. We've also provided `generate_liveness.py` and the other generation scripts to test individual runs of interference, liveness, and spill.

Finally, please don't judge my unpolished Python code, there are plenty of ways to make it better, I just needed to hack it enough to get things working *good enough* this quarter, I'll clean it up after catching up with grading.